

Data Abstraction & 3-Level Schema Architecture

Q. What is Data Abstraction in a DBMS Explain the three levels of data abstraction with a suitable example.

[DEF] Definition to Remember

Data Abstraction is the process of hiding the complex, low-level details of how data is stored and maintained, while exposing only the relevant, simplified information to end-users. It is implemented through the ANSI/SPARC three-level architecture: Physical, Logical, and View levels.

1. The Three Levels of Data Abstraction

VIEW LEVEL (External)	<- End Users see their specific views
LOGICAL LEVEL (Conceptual)	<- Designers see the full structure
PHYSICAL LEVEL (Internal)	<- DBA sees raw storage details

1. Physical Level (Internal Level) -- *Lowest Level*

2. Logical Level (Conceptual Level) -- *Middle Level*

3. View Level (External Level) -- *Highest Level*

2. Example -- University Database

Level	What is Seen
Physical Level	Records stored on Disk D in 256-byte b
Logical Level	Table `STUDENT` (Roll_No, Name, Departm
View Level (Professor)	Sees only `Roll_No` and `Name` -- fees
View Level (Accountant)	Sees only `Roll_No`, `Name`, and `Fees

[MUST-WRITE] Must-Write Points (for 10 marks)

1. Data abstraction hides complexity and shows only relevant data to each user.
2. It follows the ANSI/SPARC three-level architecture: Physical, Logical, and View.
3. Physical level deals with actual storage -- files, indexes, compression, encryption.
4. Logical level defines the full database structure: tables, columns, data types, keys.
5. View level provides user-specific views, hiding irrelevant or sensitive data.
6. Each level hides details from the level above -- enabling data independence.
7. Example: A Professor sees student names; an Accountant sees fees -- from the same database.

[RECALL] Quick Recall

Physical Level (how stored) -> Logical Level (what stored + relationships) -> View Level (user-specific view) -> Data Independence